1. Introduction

本次 Lab 5 實作三個 DQN 實驗任務：

- **Task 1**：CartPole-v1，Vanilla DQN，MLP 架構，約 **80k steps** 達到滿分 **500**，20-seed 平均 **500.00**
- **Task 2**：Pong-v5，Vanilla DQN，CNN 架構，訓練 **6M steps**，20-seed 平均 **19.40**
- **Task 3**：Pong-v5，Enhanced DQN，含 Double DQN + PER + Multi-step Return，訓練 **4M steps**，於 **~1M steps** 已穩定收斂至 ~18–20，20-seed 平均 **19.15**

Task 3 相比 Task 2，在更少的訓練步數內達到相近的最終分數，驗證三項增強技術對樣本效率的顯著提升。

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2. Implementation (20%)

2.1 How do you obtain the Bellman error for DQN?

DQN 實作 **Bellman 方程式**。每次更新從 Replay Buffer 中取樣一個 mini-batch，計算如下：

Step 1：取得當前 Q 值

```
q_values = self.q_net(states).gather(1, actions.unsqueeze(1)).squeeze(1)
```

Step 2：用 target network 計算 Bellman target

```
with torch.no_grad():
    target_q_val = rewards + self.gamma * (1 - dones) *
self.target_net(next_states).max(1)[0]
```

Step 3：計算 loss 並 gradient update

```
loss = nn.functional.mse_loss(q_values, target_q_val)
self.optimizer.zero_grad()
loss.backward()
self.optimizer.step()
```

TD error 的平方即為 Bellman Error，透過 MSE Loss 反向傳播，僅更新 online network 的參數。

每 `target_update_frequency` 步將權重完整複製到 target network：

```
if self.train_count % self.target_update_frequency == 0:
    self.target_net.load_state_dict(self.q_net.state_dict())
```

2.2 How do you modify DQN to Double DQN?

Vanilla DQN 計算 target Q 值時讓 target network 自己選動作、自己評分，容易高估。

在 `train()` 裡，我先用 `q_net` 對 next_states 取 `.argmax()` 選出它認為最好的 action，存成 `a_star`，再用 `target_net` 的 `.gather()` 去查那個 action 對應的 Q 值——選動作和評分由兩個網路分工，打破自選自評的高估問題：

```
with torch.no_grad():
    a_star = self.q_net(next_states).argmax(dim=1)
    next_q = self.target_net(next_states).gather(1,
a_star.unsqueeze(1)).squeeze(1)
    target_q = rewards + (self.gamma ** self.n_multi_step) * next_q * (1 -
dones)
```

`self.gamma ** self.n_multi_step` 是因為同時整合了 multi-step return，折扣要乘 n 次，n=3。

2.3 How do you implement the memory buffer for PER?

普通 Replay Buffer 是隨機抽樣，PER 改成 TD error 越大、sampling probability 越高，核心是三個函數：

`add`：新 transition 存入時，先給它目前 buffer 裡的最高 priority `self.max_priority`，確保每筆新資料至少被取樣過一次，之後再根據真實 TD error 更新。

```
def add(self, transition):
    if len(self.buffer) < self.capacity:
        self.buffer.append(transition)
    else:
        self.buffer[self.pos] = transition
    self.priorities[self.pos] = self.max_priority
    self.pos = (self.pos + 1) % self.capacity
```

`sample`：把 buffer 裡的 priority 除以總和轉成機率，用 `np.random.choice` 依 probability 取樣。同時每次 sample 都把 `self.beta` 往 1.0 anneal 一步，再算 IS weights 正規化後回傳：

```
def sample(self, batch_size):
    self.beta = min(1.0, self.beta + self.beta_increment)
    prios = self.priorities[:len(self.buffer)]
    probs = prios / prios.sum()
    indices = np.random.choice(len(self.buffer), batch_size, p=probs)
    samples = [self.buffer[idx] for idx in indices]
    weights = (len(self.buffer) * probs[indices]) ** (-self.beta)
    weights /= weights.max()
    return samples, indices, weights
```

`update_priorities`：每次 train 後用新算出的 TD error 更新 priority，同時追蹤 `max_priority` 供下一筆 `add` 使用：

```
def update_priorities(self, indices, errors):
    for idx, error in zip(indices, errors):
        new_prio = (error + 1e-6) ** self.alpha
        self.priorities[idx] = new_prio
        self.max_priority = max(self.max_priority, new_prio)
```

IS weights 在 `train()` 裡乘進 loss，讓被高頻取樣的 transition 不會過度主導 gradient 方向：

```
loss = (weights * nn.functional.smooth_l1_loss(
        q_values, target_q, reduction='none')).mean()
```

2.4 How do you modify the 1-step return to multi-step return?

原本 1-step 只看當步 reward 加下一步的 Q 估計；改成 n-step，n=3，就是把連續 3 步的折扣 reward 全部加起來，再接第 3 步之後的 Q 估計，讓 reward 訊號傳得更遠。

實作上在 `__init__` 建一個長度為 3 的 `multi_step_buffer`，在 `run()` 的互動迴圈裡，每走一步就 append 進去；等 buffer 滿 3 筆，取第一筆的 state/action 和最後一筆的 next_state/done 組成新 transition 存入 PER buffer：

```
self.multi_step_buffer.append((state, action, reward, next_state, done))

if len(self.multi_step_buffer) == self.n_multi_step:
    n_step_reward = sum([self.gamma**i * self.multi_step_buffer[i][2]
                         for i in range(self.n_multi_step)])
    oldest_state, oldest_action, _, _, _ = self.multi_step_buffer[0]
    _, _, _, final_next_state, final_done = self.multi_step_buffer[-1]
    self.memory.add((oldest_state, oldest_action,
```

```
                        n_step_reward, final_next_state, final_done))

if done:
    self.multi_step_buffer.clear()  # episode 結束清空，避免跨 episode 混合
```

在 `train()` 計算 target Q 時，折扣因子也要改成 `gamma ** n_multi_step`：

```
target_q = rewards + (self.gamma ** self.n_multi_step) * next_q * (1 -
dones)
```

2.5 Explain how you use Weights & Biases to track the model performance.

在 `__main__` 最後呼叫 `wandb.init()` 綁定 project 與 run 名稱：

```
wandb.init(project="DLP-Lab5-DQN-Pong_tk3", name=args.wandb_run_name,
save_code=True)
```

訓練過程中分三個地方 log：每 1000 env steps log 一次訓練狀態、每個 episode 結束 log 該 episode 的 total reward、每 10 episodes 的 evaluation 結果 log eval reward：

```
# 每 1000 env steps
wandb.log({"Env Step Count": self.env_count, "Epsilon": self.epsilon,
...})

# 每個 episode 結束
wandb.log({"Total Reward": total_reward, "Env Step Count": self.env_count,
...})

# 每 10 episodes evaluation
wandb.log({"Eval Reward": eval_reward, "Env Step Count": self.env_count})

# train() 裡每次 update
wandb.log({"Loss": loss.item(), "Env Step Count": self.env_count})
```

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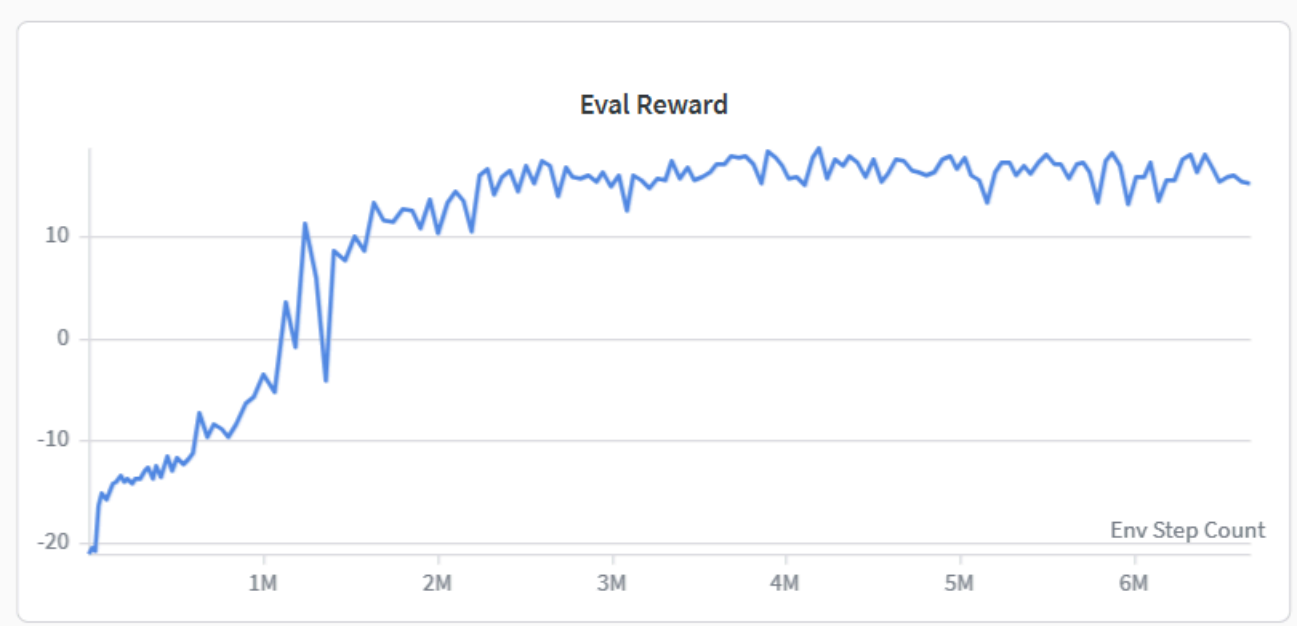
3. Analysis and Discussions (25%)

3.1 Training Curves

Task 1 — CartPole-v1

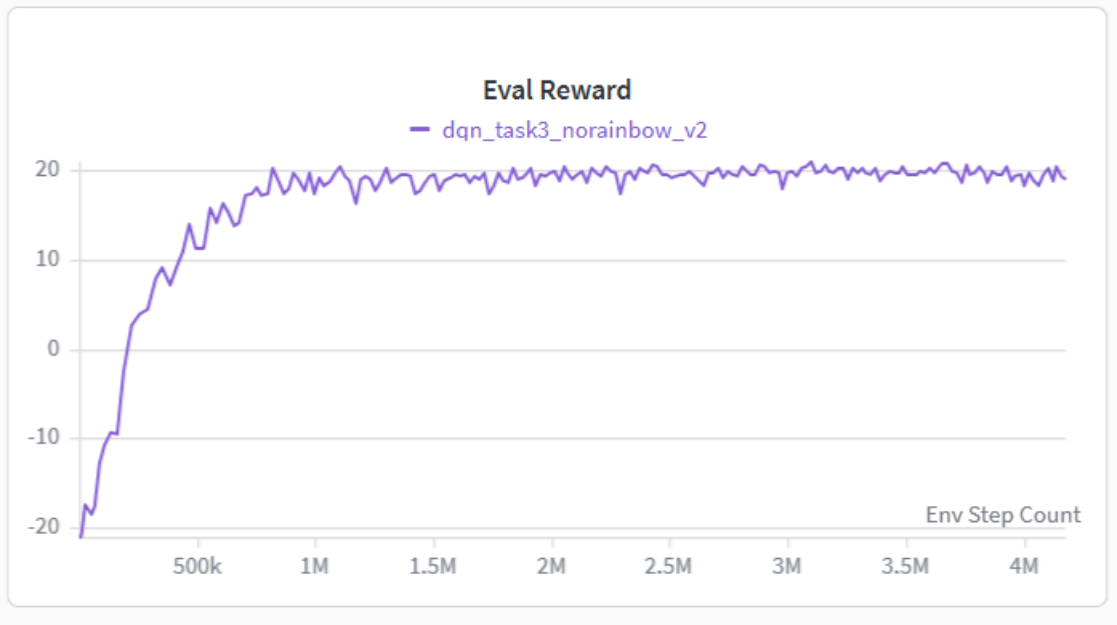
約在 **80k steps** 時 eval reward 突破 480。

Task 2 — Pong-v5，Vanilla DQN



從初始 -20 起步，約在 **2M steps** 進入穩定上升。

Task 3 — Pong-v5，Enhanced DQN



Enhanced DQN 同樣從 -20 起步，但約在 **500k steps** 已可看到明顯上升趨勢，至 **1M steps** 時 eval reward 穩定進入 ~18–20 區間。

3.2 Sample Efficiency Analysis

	Task 1	Task 2	Task 3
穩定收斂	~80k steps	~2M steps	~1M steps
狀態空間	4 維連續向量	4×84×84 像素	4×84×84 像素
網路架構	MLP	CNN	Dueling CNN
Reward 訊號	每步 +1	得/失分才 ±1	同 Task 2
20-seed 測試平均分	**500.00 / 500**	**19.40 / 21**	**19.15 / 21**

Conclusion：

1. **Task 1**：CartPole 環境狀態為 4 個連續數值，MLP 易於學習；reward 每存活一步即得 +1，約 80k steps 達到滿分後長期穩定。
2. **Task 2**：Pong 輸入為高維像素畫面 84×84×4，CNN 需要更多資料才能抽取有效視覺特徵；reward 僅在得分或失分時才有 ±1 訊號，導致 TD error 在早期幾乎無有效訊息，需 2M+ steps 才開始收斂，6M steps 方達穩定。
3. **Task 3**：Enhanced DQN 在 ~1M steps 即穩定收斂至 ~18–20 eval reward，遠早於 Vanilla DQN 的 2M+ steps，且最終 20-seed 平均分 19.15 與 Vanilla DQN 的 19.40 相當，顯示三項增強技術大幅提升樣本效率而不犧牲最終表現。

3.3 Ablation Study

本實驗以 **Task 2 Vanilla DQN** 作為 baseline，對比 **Task 3 Enhanced DQN**，含 DDQN + PER + Multi-step 全啟用。以下結合理論分析與整合實驗結果說明各技術的貢獻：

技術	解決的核心問題	預期效益
Double DQN	Vanilla DQN 高估 Q 值	抑制 Q 值高估，訓練更穩定
PER	忽視高 TD error 的稀有得分 transition	加速早期學習，reward 利用效率提升
Multi-step Return	1-step return bias 大，reward 傳遞緩慢	降低 bootstrap bias，reward 能跨多步回溯，加速收斂

效果對比：

	Task 2	Task 3
訓練總 steps	~6M steps	~4M steps
穩定收斂 steps	~2M steps	**~1M steps**
20-seed 測試平均	**19.40**	**19.15**

Conclusion：三項技巧能夠大幅減少訓練步數，達到與 Vanilla DQN 相近的最終表現。PER 讓 TD error 過高的樣本以更高 probability 被取樣；Multi-step Return 讓 reward 能回溯影響前 n 步的 Q 值；Double DQN 確保 Q 值估計保守準確，防止高 probability 取樣的樣本因 Q 值高估而污染 gradient。

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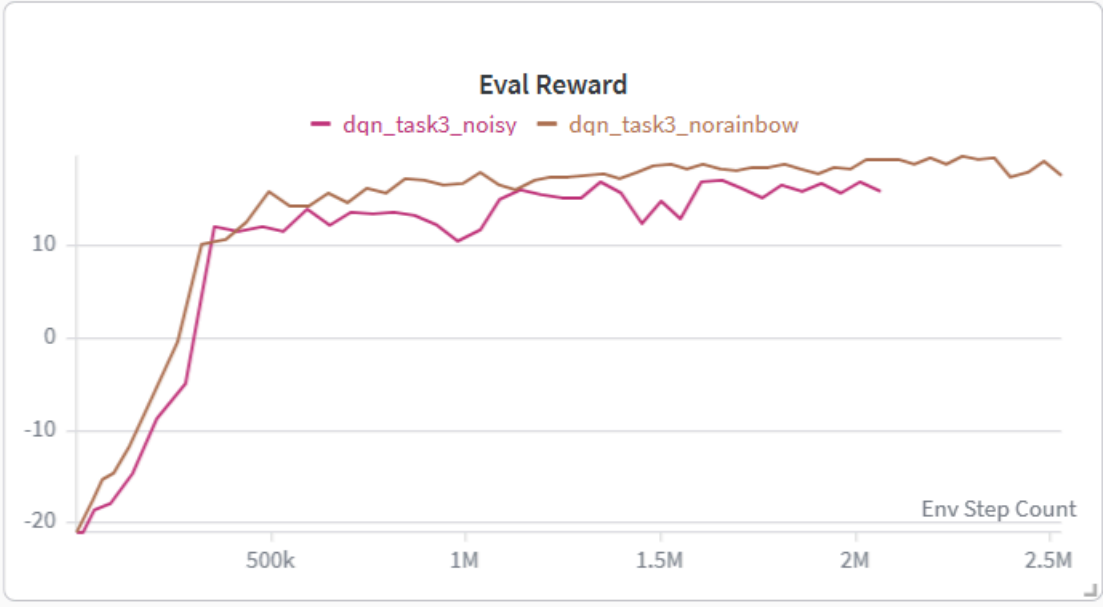
4. Additional Analysis (Bonus up to 10%)

4.1 Noisy Linear vs. ε-greedy Exploration

本節比較兩種 exploration 策略的訓練效率：

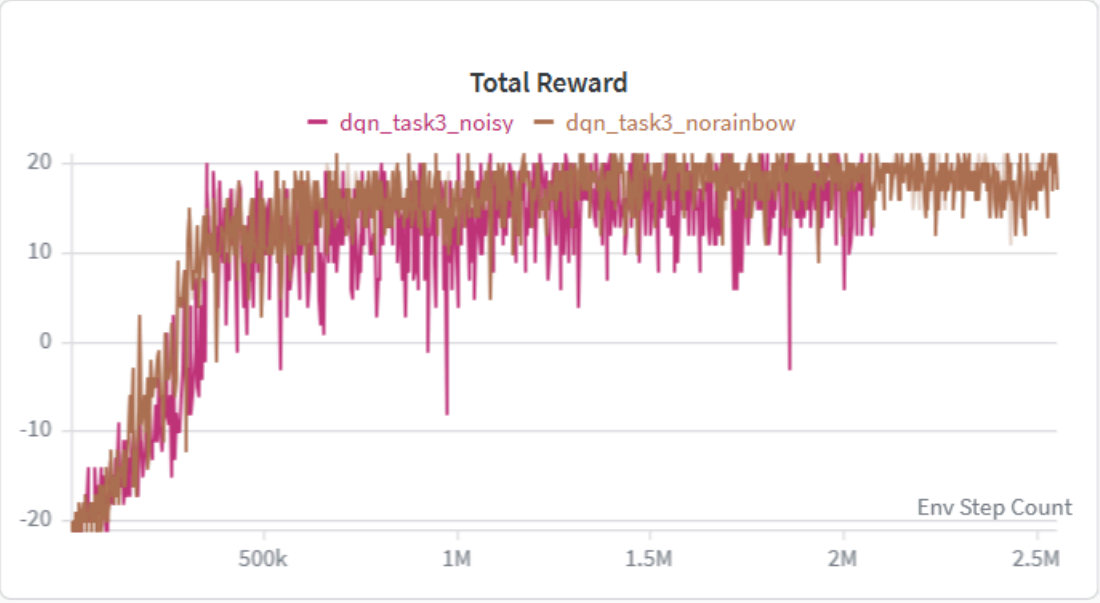
- `dqn_task3_norainbow`：使用傳統 **ε-greedy** 策略，ε 從 1.0 線性衰減至 0.01，搭配 DDQN + PER + Multi-step Return
- `dqn_task3_noisy`：在 Task 3 基礎上以 **Noisy Linear Layer** 取代 ε-greedy；網路權重本身帶有可學習的隨機雜訊 μ + σ·ε，在推論時自動進行探索，不需手動設定 ε 衰減

Eval Reward 比較



上圖為兩個 run 的 Eval Reward，每 10 episodes 評估一次，隨 Env Steps 的變化。此比較基於 v1 版本的訓練結果：`dqn_task3_norainbow` 橘色整體曲線較為平滑，在 2M steps 後穩定維持於約 **13–15**；`dqn_task3_noisy` 粉紅色呈現較高的震盪，同樣最終收斂至 ~13–15，但 variance 明顯更大。

Total Reward 比較



上圖為每個 episode 的 Total Reward 趨勢。兩個 run 在 2.5M steps 附近均趨近於 ~20，但 noisy 版本的波動幅度更劇烈，norainbow 版本則相對穩定地收斂。

結論

面向	ε-greedy / norainbow	Noisy Linear / noisy
探索方式	ε 線性衰減	隨學習自動調整
Eval Reward 穩定性	曲線平滑	震盪幅度大
收斂速度	約 500k steps 開始上升	約 500k steps 開始上升，速度相近

面向	ε-greedy / norainbow	Noisy Linear / noisy
最終 eval 平均	~18–20	~13–15

Noisy Linear 的優點在於不需要人工設計 ε-decay 排程；但在 Pong 這種環境下，高 variance 特性可能反而造成訓練不穩定。ε-greedy 搭配線性衰減在本實驗中顯示出更一致的收斂行為。

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5. Instruction

5.1 Train

Task 1 — CartPole-v1

```
python dqn_task1.py
```

Task 2 — Pong-v5

```
python dqn_task2.py
```

Task 3 — Pong-v5 Enhanced

```
python dqn_task3.py \
    --epsilon-decay-steps 80000 \
    --lr 5e-5 \
    --target-update-frequency 1000 \
    --replay-start-size 5000
```

5.2 Test

Task 1 — seed 314553044 ~ 314553063，共 20 個

```
python .\test_model12.py --model_path ..\LAB5_314553044_task1.pt --seed 0
```

Task 2 — seed 314553044 ~ 314553063，共 20 個

```
python .\test_model12.py --model_path ..\LAB5_314553044_task2.pt --seed 0
--env-name ALE/Pong-v5
```

Task 3 best — seed 0 ~ 19，共 20 個
(使用**969174 env steps**的權重)

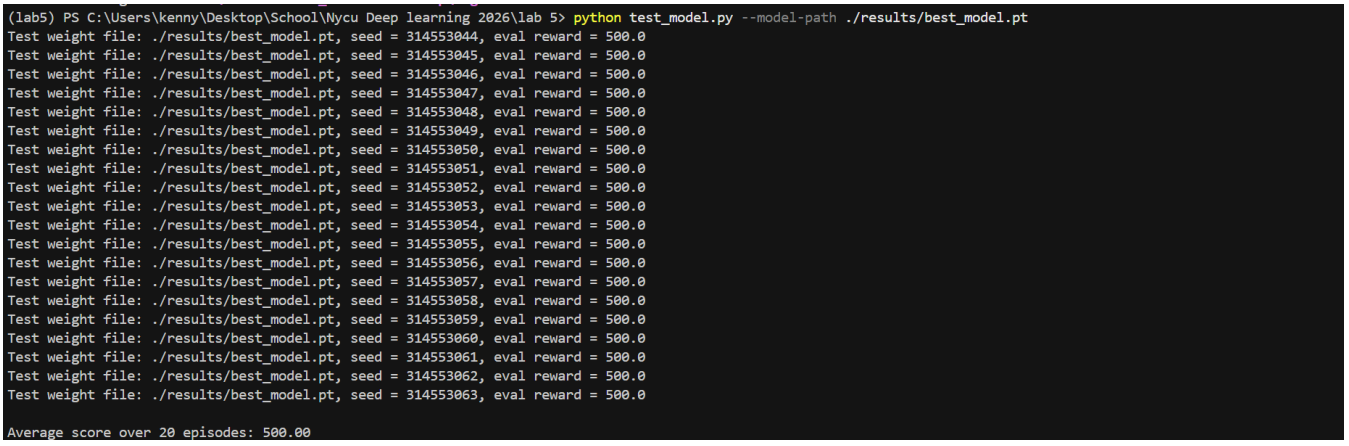
```
 python .\test_model3.py --model_path ..\LAB5_314553044_task3_best.pt --
seed 0
```

Task 3 Milestone snapshots — 各 checkpoint 評估

```
 python .\test_model3.py --model_path ..\LAB5_314553044_task3_600000.pt --
seed 0

 python .\test_model3.py --model_path ..\LAB5_314553044_task3_1000000.pt -
-seed 0

 python .\test_model3.py --model_path ..\LAB5_314553044_task3_1500000.pt -
-seed 0

 python .\test_model3.py --model_path ..\LAB5_314553044_task3_2000000.pt -
-seed 0

python test_model3_no_rainbow.py \
 python .\test_model3.py --model_path ..\LAB5_314553044_task3_2500000.pt -
-seed 0
```

5.3 Test Result

Task 1



```
(lab5) PS C:\Users\kenny\Desktop\School\Nycu Deep learning 2026\lab 5> python test_model.py --model-path ./results/best_model.pt
Test weight file: ./results/best_model.pt, seed = 314553044, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553045, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553046, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553047, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553048, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553049, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553050, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553051, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553052, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553053, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553054, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553055, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553056, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553057, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553058, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553059, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553060, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553061, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553062, eval reward = 500.0
Test weight file: ./results/best_model.pt, seed = 314553063, eval reward = 500.0

Average score over 20 episodes: 500.00
```

Average score over 20 episodes: 500.00

Task 2

```
(lab5) PS C:\Users\kenny\Desktop\School\Nycu Deep learning 2026\lab 5> python .\test_model12.py --model-path .\results\task2\LAB5_314553044_task2.pt --env-name ALE/Pong
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553044, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553045, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553046, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553047, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553048, eval reward = 21.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553049, eval reward = 17.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553050, eval reward = 21.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553051, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553052, eval reward = 21.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553053, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553054, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553055, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553056, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553057, eval reward = 17.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553058, eval reward = 17.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553059, eval reward = 21.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553060, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553061, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553062, eval reward = 21.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 314553063, eval reward = 20.0

Average score over 20 episodes: 19.40
```

Average score over 20 episodes: 19.40

```
(lab5) PS C:\Users\kenny\Desktop\School\Nycu Deep learning 2026\lab 5> python .\test_model12.py --model-path .\results\task2\LAB5_314553044_task2.pt --env-name ALE/Pong-v5
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 0, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 1, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 2, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 3, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 4, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 5, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 6, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 7, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 8, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 9, eval reward = 17.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 10, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 11, eval reward = 19.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 12, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 13, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 14, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 15, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 16, eval reward = 18.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 17, eval reward = 17.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 18, eval reward = 20.0
Test weight file: .\results\task2\LAB5_314553044_task2.pt, seed = 19, eval reward = 19.0

Average score over 20 episodes: 18.75
```

Average score over 20 episodes: 18.75

Task 3

```
(lab5-env) kenny@instance-20260502-083543:~$ python test_model3_no_rainbow.py --model_path results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt  --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 0, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 1, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 2, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 3, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 4, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 5, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 6, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 7, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 8, eval reward = 17.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 9, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 10, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 11, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 12, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 13, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 14, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 15, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 16, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 17, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 18, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_step969174_score19.8.pt, seed = 19, eval reward = 19.0

Average score over 20 episodes: 19.15
```

Average score over 20 episodes: 19.15

使用**969174 env steps**的權重

600,000 env steps

```
(lab5-env) kenny@instance-20260502-083543:~$ python test_model3_no_rainbow.py --model_path results_norainbow_v2_/LAB5_314553044_task3_600000.pt  --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 0, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 1, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 2, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 3, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 4, eval reward = 11.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 5, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 6, eval reward = 14.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 7, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 8, eval reward = 17.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 9, eval reward = 17.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 10, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 11, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 12, eval reward = 15.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 13, eval reward = 14.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 14, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 15, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 16, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 17, eval reward = 10.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 18, eval reward = 13.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_600000.pt, seed = 19, eval reward = 15.0

Average score over 20 episodes: 14.60
```

1,000,000 env steps

```
(lab5-env) kenny@instance-20260502-083543:~$ python test_model3_no_rainbow.py --model_path results_norainbow_v2_/LAB5_314553044_task3_1000000.pt  --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 0, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 1, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 2, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 3, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 4, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 5, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 6, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 7, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 8, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 9, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 10, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 11, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 12, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 13, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 14, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 15, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 16, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 17, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 18, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1000000.pt, seed = 19, eval reward = 18.0

Average score over 20 episodes: 19.10
```

1,500,000 env steps

```
(lab5-env) kenny@instance-20260502-083543:~$ python test_model3_no_rainbow.py --model_path results_norainbow_v2_/LAB5_314553044_task3_1500000.pt  --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 0, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 1, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 2, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 3, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 4, eval reward = 16.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 5, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 6, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 7, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 8, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 9, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 10, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 11, eval reward = 21.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 12, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 13, eval reward = 19.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 14, eval reward = 18.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 15, eval reward = 15.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 16, eval reward = 17.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 17, eval reward = 17.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 18, eval reward = 20.0
Test weight file: results_norainbow_v2_/LAB5_314553044_task3_1500000.pt, seed = 19, eval reward = 20.0

Average score over 20 episodes: 18.90
```

2,000,000 env steps

```
(lab5) PS C:\Users\kenny\Desktop\School\Nycu Deep learning 2026\lab 5> python .\test_model3.py --model_path .\results\task3\LAB5_314553044_task3_2000000.pt --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 0, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 1, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 2, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 3, eval reward = 13.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 4, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 5, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 6, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 7, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 8, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 9, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 10, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 11, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 12, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 13, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 14, eval reward = 17.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 15, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 16, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 17, eval reward = 14.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 18, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2000000.pt, seed = 19, eval reward = 16.0

Average score over 20 episodes: 18.65
```

2,500,000 env steps

```
(lab5) PS C:\Users\kenny\Desktop\School\Nycu Deep learning 2026\lab 5> python .\test_model3.py --model_path .\results\task3\LAB5_314553044_task3_2500000.pt --seed 0
A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)
[Powered by Stella]
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 0, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 1, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 2, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 3, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 4, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 5, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 6, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 7, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 8, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 9, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 10, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 11, eval reward = 19.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 12, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 13, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 14, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 15, eval reward = 14.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 16, eval reward = 21.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 17, eval reward = 18.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 18, eval reward = 20.0
Test weight file: .\results\task3\LAB5_314553044_task3_2500000.pt, seed = 19, eval reward = 15.0

Average score over 20 episodes: 19.00
```

5.4 Parameter

Task 1 — CartPole-v1

Hyperparameter	Value
batch_size	64
memory_size	100,000
learning_rate	0.0001
discount_factor γ	0.99
num_episodes	5,000
epsilon_start	1.0
epsilon_min	0.01
epsilon_decay_steps	1,000,000
target_update_frequency	4,000 steps
replay_start_size	2,000
max_episode_steps	10,000
train_per_step	1

Hyperparameter	Value
Network	Linear(4→128)→ReLU→Linear(128→128)→ReLU→Linear(128→2)

Task 2 — Pong-v5

Hyperparameter	Value
batch_size	64
memory_size	500,000
learning_rate	0.0001
discount_factor γ	0.99
num_episodes	5,000
epsilon_start	1.0
epsilon_min	0.01
epsilon_decay_steps	250,000
target_update_frequency	1,000 steps
replay_start_size	5,000
max_episode_steps	5,000
train_per_step	1
Network	Conv2d(4,32,8,4)→Conv2d(32,64,4,2)→Conv2d(64,64,3,1)→Linear(3136,512)→Linear(512,6)

Task 3 — Pong-v5 Enhanced

Hyperparameter	Value
batch_size	64
memory_size	500,000
learning_rate	5e-05
discount_factor γ	0.99
num_episodes	5,000
epsilon_start	1.0
epsilon_min	0.01
epsilon_decay_steps	80,000
target_update_frequency	1,000 steps
replay_start_size	5,000
max_episode_steps	5,000

Hyperparameter	Value
train_per_step	1
training seed	42
Enhancements	Double DQN + PER + Multi-step Return, n=3
Network	同 Task 2 架構，加入 Dueling DQN